सी डैक | प्रगत संगणन विकास केंद्र, नोएडा
CDAC | Centre for Development of Advanced Computing, Noida

A PROJECT REPORT

ON

Sentiment Analysis for Depression Using Machine Learning & NLP

Submitted In Partial Fulfilment of the Requirement for the Award of

Post Graduate Diploma in Artificial Intelligence (PG-DAI)

Under the Guidance of

MR. RAVI PAYAL
(Joint Director)
(Course Co-ordinator)

MRS. SIDDHIDATRI NAYAK
(Project Guide)

Submitted By

NIVEDITA A KALEL
PRN: 220320528045

SAFIYA SAYYAD
PRN: 220320528001

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CDAC, B-30, Institutional Area, Sector-62

Noida (Uttar Pradesh)-201307

CERTIFICATE

CDAC, NOIDA

This is to certify that Report entitled **"Sentiment Analysis for Depression Using Machine Learning & NLP"** which is submitted by Nivedita Kalel and Safiya Sayyad in partial fulfilment of the requirement for the award of **Post Graduate Diploma in Artificial Intelligence** (PG-DAI) to **CDAC, Noida is** a record of the candidates own work carried out by them under my supervision.

The documentation embodies results of original work, and studies are carried out by the student themselves and the contents of the report do not from the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

Mrs. SIDDHIDATRI NAYAK

(Project Guide)

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NIVEDITA A KALEL

PRN: 220320528045

SAFIYA SAYYAD

PRN: 220320528001

ABSTRACT

Today the different social networking sites have enabled everyone to easily express and share their feelings with people around the world. A lot of people use text for communicating, which can be done through different social media messaging platforms available today such as Twitter, Facebook etc, as they find it easier to express their feelings through text instead of speaking them out. Many people who also suffer from stress find it easier to express their feelings on online platform, as over there they can express themselves very easily. So if they are alerted beforehand, there are ways to overcome the mental problems and stress they are suffering from. Depression stands out to be one of the most well-known mental health disorders and a major issue for medical and mental health practitioners. Legitimate checking can help in its discovery, which could be useful to anticipate and prevent depression all-together. Hence there is a need for a system, which can cater to such issues and help the user. The purpose of this paper is to propose an efficient method that can detect the level of depression in Twitter users. Sentiment scores calculated can be combined with different emotions to provide a better method to calculate depression scores. This process will help underscore various aspects of depression that have not been understood previously. The main aim is to provide a sense of understanding regarding depression levels in different users and how the scores can be correlated to the main data.

INTRODUCTION TO THE PROBLEM STATEMENT AND THE POSSIBLE SOLUTION

Sentiment analysis refers to identifying as well as classifying the sentiments that are expressed in the text source. Tweets are often useful in generating a vast amount of sentiment data upon analysis. These data are useful in understanding the opinion of the people.

Therefore we need to develop an Automated Machine Learning Sentiment Analysis Model in order to compute the customer perception. Due to the presence of non-useful characters (collectively termed as the noise) along with useful data, it becomes difficult to implement models on them.

In this project, we aim to analyze the sentiment of the tweets provided from the twitter dataset by developing a machine learning model involving the use of classifiers (**XGboost Classifier, Logistic Regression, SVM, Random Forest Classifier and Decision Tree**) The performance of these classifiers is then evaluated using accuracy .

DATASET DESCRIPTION

In this project, we try to implement a Twitter sentiment analysis model that helps to overcome the challenges of identifying the sentiments of the tweets. The necessary details regarding the dataset are:

The dataset provided is the **Twitter Dataset** which consists of **70,000** tweets that have been extracted using the Twitter API. The various columns present in the dataset are:

- target: the polarity of the tweet (positive or negative)
- ids: Unique id of the tweet
- date: the date of the tweet
- flag: It refers to the query. If no such query exists then it is NO QUERY.
- user: It refers to the name of the user that tweeted
- text: It refers to the text of the tweet

HOW DOES SOCIAL MEDIA PLAY A ROLE IN DEPRESSION?

Depression is a mental illness that is not taken seriously in some countries. Some people have been diagnosed with depression and don't take medication to overcome it. They just continue their lives without medication and getting worse.

There are many experiences that can cause us depression, such losing a loved one, losing a job, getting a divorce and other tough situations can lead a person to feel blue, lonely and overwhelmed. It's normal to get these feelings when we feel stressed. Everybody have experimented sadness at times. However, depression is very different from this. Depression is a psychiatric disorder that need to be addressed with medication. Definition of Depression.

According to Our World in Data Website, Depressive disorders occur with varying severity. The WHO's International Classification of Diseases (ICD-10) define this set of disorders ranging from mild to moderate to severe. The Institute for Health Metrics and Evaluation (IHME) adopt such definitions by disaggregating to mild, persistent depression (dysthymia) and major depressive disorder (severe).

All forms of depressive disorder experience some of the following symptoms:

(a) Reduced concentration and attention
(b) Reduced self-esteem and self-confidence
(c) Ideas of guilt and unworthiness (even in a mild type of episode)
(d) Bleak and pessimistic views of the future
(e) Ideas or acts of self-harm or suicide
(f) Disturbed sleep
(g) Diminished appetite

Worldwide Statistics

322 million people worldwide live with depression.

Depression is the leading cause of disability worldwide. Almost 75% of people with mental disorders remain untreated in developing countries with almost 1 million people taking their lives each year. In addition, according to the World

Health Organization (WHO), 1 in 13 globally suffers from anxiety. The WHO

reports that anxiety disorders are the most common mental disorders worldwide with specific phobia, major depressive disorder and social phobia being the most common anxiety disorders. Depression on Social Media
Social media platforms are becoming an integral part of people's life. They reflect user's personal life. People likes to share happiness, joy and sadness on social media. These platforms are used for researchers to identify the causes of depression and detect it.

Detecting earlier depression can be a huge step to address the mental illness and offer support to the people suffering from this terrible mental illness. Creating a model to detect depression in tweets.

In Machine Learning, there are many ways for sentiment analysis such: decision-based systems, Bayesian classifiers, support vector machine, neural networks and sample-based methods.

DATA PRE-PROCESSING

In the above-given problem statement before training the model, we have performed various pre-processing steps on the dataset that mainly dealt with removing stop words, removing emoji's. The text document is then converted into the lowercase for better generalization.

Subsequently, the punctuations were cleaned and removed thereby reducing the unnecessary noise from the dataset. After that, we have also removed the repeating characters from the words along with removing the URLs as they do not have any significant importance.

At last, we then performed Stemming (reducing the words to their derived stems) and Lemmatization (reducing the derived words to their root form known as lemma) for better results.

MODEL BUILDING

After training the model we then apply the evaluation measures to check how the model is performing. Accordingly, we use the accuracy score to check the performance of the models respectively.

In the problem statement we have used three different models respectively:

- XG Boost Classifier
- Random Forest Classifier
- Logistic Regression
- SVM (Support Vector Machine)
- Decision tree

The idea behind choosing these models is that we want to try all the classifiers on the dataset ranging from simple ones to complex models and then try to find out the one which gives the best performance among them.

CODING

```
import re    # for regular expressions
import nltk  # for text manipulation
import string # for text manipulation
import warnings
import numpy as np
import pandas as pd # for data manipulation
import matplotlib.pyplot as plt

pd.set_option("display.max_colwidth", 200)
warnings.filterwarnings("ignore") #ignore warnings

import os
for dirname, _, filenames in os.walk('Project.csv'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

%matplotlib inline

data = pd.read_csv("../input/dataset1/Project.csv",encoding='latin-1')
data.head()

print('Length of the dataset is:',len(data))
data.shape
```

```
DATASET_COLUMNS = ["target", "ids", "date", "flag", "user", "TweetText"]
data.columns = DATASET_COLUMNS
data.head()

data.info()

data.isnull().sum()

data.drop(['ids','date','flag','user'],axis = 1,inplace = True)

data.head()
```

Data Cleaning

Removing Twitter Handles (@user)

```
data['Clean_TweetText'] = data['TweetText'].str.replace("@", "")
data.head()
```

Removing links

```
data['Clean_TweetText'] = data['Clean_TweetText'].str.replace(r"http\S+", "")
data.head()
```

Removing Punctuations, Numbers, and Special Characters

```
data['Clean_TweetText'] = data['Clean_TweetText'].str.replace("[^a-zA-Z]", " ")
data.head()
```

```
data['target'] = data['target'].replace(4,1)

data['target'].unique()
```

Remove Stop Words

```
import nltk
nltk.download('stopwords')
stopwords=nltk.corpus.stopwords.words('english')

def remove_stopwords(text):
    clean_text=' '.join([word for word in text.split() if word not in stopwords])
    return clean_text

data['Clean_TweetText'] = data['Clean_TweetText'].apply(lambda text :
remove_stopwords(text.lower()))
data.head()
```

Text Tokenization and Normalization

```
data['Clean_TweetText'] = data['Clean_TweetText'].apply(lambda x: x.split())
data.head()

from nltk.stem.porter import *
stemmer = PorterStemmer()
data['Clean_TweetText'] = data['Clean_TweetText'].apply(lambda x:
[stemmer.stem(i) for i in x])
data.head()
```

Now let’s stitch these tokens back together

```
data['Clean_TweetText'] = data['Clean_TweetText'].apply(lambda x: ' '.join([w for w in x]))
data.head()
```

Removing small words

```
data['Clean_TweetText'] = data['Clean_TweetText'].apply(lambda x: ' '.join([w for w in x.split() if len(w)>3]))
data.head()
data.shape
```

```
from wordcloud import WordCloud
```

Data Vizualisation

In this section we will visualize the tweets using wordclouds. A wordcloud is a visualization wherein the most frequent words appear in large size and the less frequent words appear in smaller sizes.

What are the most common words in the entire dataset?

```
all_words = data['Clean_TweetText']
wc = WordCloud(max_words = 1000 , width = 1600 , height = 800,
            collocations=False).generate(" ".join(all_words))
plt.figure(figsize = (20,20))
plt.imshow(wc)
all_words.shape
```

What are the most common words in the dataset for Positive and Depressive tweets, respectively?

```
data_neg = data['Clean_TweetText'][:35001]
```

```
plt.figure(figsize = (20,20))
wc = WordCloud(max_words = 1000 , width = 1600 , height = 800,
            collocations=False).generate(" ".join(data_neg))
plt.imshow(wc)
data_neg.shape

data_pos = data['Clean_TweetText'][35001:]
wc = WordCloud(max_words = 1000 , width = 1600 , height = 800,
            collocations=False).generate(" ".join(data_pos))
plt.figure(figsize = (20,20))
plt.imshow(wc)
data_pos.shape
```

Model building

```
pip install xgboost

from xgboost import XGBClassifier
import xgboost as xgb
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn import svm
from sklearn.metrics import accuracy_score, confusion_matrix,
ConfusionMatrixDisplay
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer

count_vectorizer = CountVectorizer(stop_words='english')
cv = count_vectorizer.fit_transform(data['Clean_TweetText'])
cv.shape
```

```
X_train,X_test,y_train,y_test = train_test_split(cv,data['target'] ,
test_size=.2,stratify=data['target'], random_state=40)

import matplotlib.pyplot as plt
import seaborn as sns
```

XG-BOOST CLASSIFIER

```
xgbc = XGBClassifier(max_depth=6, n_estimators=500, nthread= 3)
xgbc.fit(X_train,y_train)
xgb_pred_train = xgbc.predict(X_train)
xgb_pred_test = xgbc.predict(X_test)
print('Training accuracy:',accuracy_score(xgb_pred_train,y_train))
print('Testing accuracy:',accuracy_score(xgb_pred_test,y_test))
```

#Confusion matrix

```
cm = confusion_matrix(y_test, xgb_pred_test)
display_labels=['True Neg','False Pos', 'False Neg','True Pos']
class_names=['positive','negative']
disp = ConfusionMatrixDisplay(confusion_matrix=cm,display_labels =
class_names)

fig, ax = plt.subplots(figsize = (9,9))
disp.plot(ax = ax, cmap=plt.cm.Blues,include_values=False);
ax.set_xticklabels(class_names,rotation=45, ha='right');
```

RANDOM FOREST CLASSIFIER

```
rf = RandomForestClassifier(n_estimators=500, random_state=42)
```

```
rf.fit(X_train,y_train)
predict_rf_train = rf.predict(X_train)
predict_rf_test = rf.predict(X_test)
print('Training accuracy:',accuracy_score(predict_rf_train,y_train))
print('Testing accuracy:',accuracy_score(predict_rf_test,y_test))
```

#Confusion matrix

```
cm = confusion_matrix(y_test, predict_rf_test)
display_labels=['True Neg','False Pos', 'False Neg','True Pos']
class_names=['positive','negative']
disp = ConfusionMatrixDisplay(confusion_matrix=cm,display_labels =
class_names)

fig, ax = plt.subplots(figsize = (9,9))
disp.plot(ax = ax, cmap=plt.cm.Blues,include_values=False);
ax.set_xticklabels(class_names,rotation=45, ha='right');
```

LOGISTIC REGRESSION

```
lr = LogisticRegression()
lr.fit(X_train,y_train)
predict_lr_train = lr.predict(X_train)
predict_lr_test = lr.predict(X_test)
print('Training accuracy:',accuracy_score(predict_lr_train,y_train))
print('Testing accuracy:',accuracy_score(predict_lr_test,y_test))
```

#Confusion matrix

```
cm = confusion_matrix(y_test, predict_lr_test)
display_labels=['True Neg','False Pos', 'False Neg','True Pos']
class_names=['positive','negative']
disp = ConfusionMatrixDisplay(confusion_matrix=cm,display_labels =
class_names)

fig, ax = plt.subplots(figsize = (9,9))
disp.plot(ax = ax, cmap=plt.cm.Blues,include_values=False);
ax.set_xticklabels(class_names,rotation=45, ha='right');
```

SUPPORT VECTOR MACHINE

```
svc = svm.SVC()
svc.fit(X_train,y_train)
predict_svc_train = svc.predict(X_train)
predict_svc_test = svc.predict(X_test)
print('Training accuracy:',accuracy_score(predict_svc_train,y_train))
print('Testing accuracy:',accuracy_score(predict_svc_test,y_test))
```

#Confusion matrix

```
cm = confusion_matrix(y_test, predict_svc_test)
display_labels=['True Neg','False Pos', 'False Neg','True Pos']
class_names=['positive','negative']
disp = ConfusionMatrixDisplay(confusion_matrix=cm,display_labels =
class_names)

fig, ax = plt.subplots(figsize = (9,9))
disp.plot(ax = ax, cmap=plt.cm.Blues,include_values=False);
ax.set_xticklabels(class_names,rotation=45, ha='right');
```

DECISION TREE

```
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(criterion = 'entropy')
dt.fit(X_train, y_train)
y_pred_train = dt.predict(X_train)
y_pred_test = dt.predict(X_test)
print('Training accuracy:',accuracy_score(y_train,y_pred_train))
print('Testing accuracy:',accuracy_score(y_test,y_pred_test))
```

#Confusion matrix

```
cm = confusion_matrix(y_test, y_pred_test)
display_labels=['True Neg','False Pos', 'False Neg','True Pos']
class_names=['positive','negative']
disp = ConfusionMatrixDisplay(confusion_matrix=cm,display_labels =
class_names)

fig, ax = plt.subplots(figsize = (9,9))
disp.plot(ax = ax, cmap=plt.cm.Blues,include_values=False);
ax.set_xticklabels(class_names,rotation=45, ha='right');
```

DATA VISUALISATION FOR SENTIMENTS

```
import seaborn as sns
sns.countplot(x='target', data=data)
```

SENTIMENT ANALYSIS WEBPAGE

app.py

```python
import ...

nltk.download('stopwords')

set(stopwords.words('english'))

app = Flask(__name__)

@app.route('/')
def my_form():
    return render_template('form.html')

@app.route('/', methods=['POST'])
def my_form_post():
    stop_words = stopwords.words('english')

    #convert to lowercase
    text1 = request.form['text1'].lower()

    text_final = ''.join(c for c in text1 if not c.isdigit())

    #remove punctuations
    #text3 = ''.join(c for c in text2 if c not in punctuation)

    #remove stopwords
    processed_doc1 = ' '.join([word for word in text_final.split() if word not in stop_words])

    sa = SentimentIntensityAnalyzer()
    dd = sa.polarity_scores(text=processed_doc1)
```

```python
    sa = SentimentIntensityAnalyzer()
    dd = sa.polarity_scores(text=processed_doc1)
    compound = round((1 + dd['compound'])/2, 2)

    return render_template('form.html', final=compound, text1=text_final,text2=dd['pos'],text5=dd['neg'],text4=compound,text3=dd['neu'])

if __name__ == "__main__":
    app.run(debug=True, host="127.0.0.1", port=5002, threaded=True)
```

Commands to run the Code: =

```
C:\Windows\System32\cmd.exe - python app.py
(c) Microsoft Corporation. All rights reserved.

E:\study\Tem_Desktop_Data\Project\API\Sentiment-Analysis-ML-Flask-App>python -m pip install --user -r requirements.txt
Collecting click==6.7
  Using cached click-6.7-py2.py3-none-any.whl (71 kB)
Collecting Flask==1.0.2
  Using cached Flask-1.0.2-py2.py3-none-any.whl (91 kB)
Collecting Flask-WTF==0.14.2
  Using cached Flask_WTF-0.14.2-py2.py3-none-any.whl (14 kB)
Collecting gunicorn==19.9.0
  Using cached gunicorn-19.9.0-py2.py3-none-any.whl (112 kB)
Collecting itsdangerous==0.24
  Using cached itsdangerous-0.24.tar.gz (46 kB)
  Preparing metadata (setup.py) ... done
Collecting Jinja2==2.10
  Using cached Jinja2-2.10-py2.py3-none-any.whl (126 kB)
Collecting MarkupSafe==1.0
  Using cached MarkupSafe-1.0.tar.gz (14 kB)
```

```
E:\study\Tem_Desktop_Data\Project\API\Sentiment-Analysis-ML-Flask-App>python app.py
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\NIVI\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
 * Serving Flask app 'app'
 * Debug mode: on
```

Generation of the link to the webpage:

```
 * Running on http://127.0.0.1:5002
Press CTRL+C to quit
 * Restarting with stat
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\NIVI\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
 * Debugger is active!
 * Debugger PIN: 109-435-784
127.0.0.1 - - [28/Sep/2022 12:01:07] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [28/Sep/2022 12:01:13] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [28/Sep/2022 12:01:27] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [28/Sep/2022 12:02:45] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [28/Sep/2022 12:02:55] "POST / HTTP/1.1" 200 -
```

RESULTS

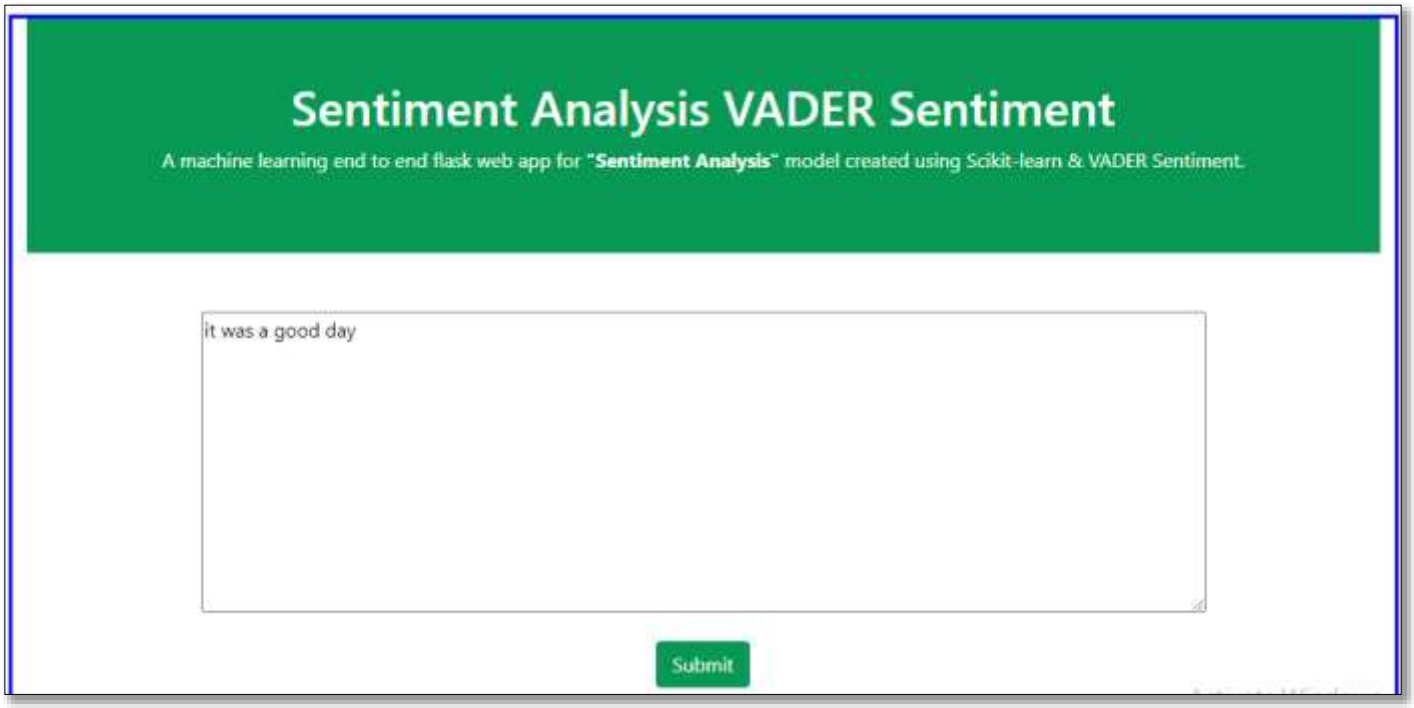
This is the result table that shows the comparison of different algorithms which are used for classification of sentiments:

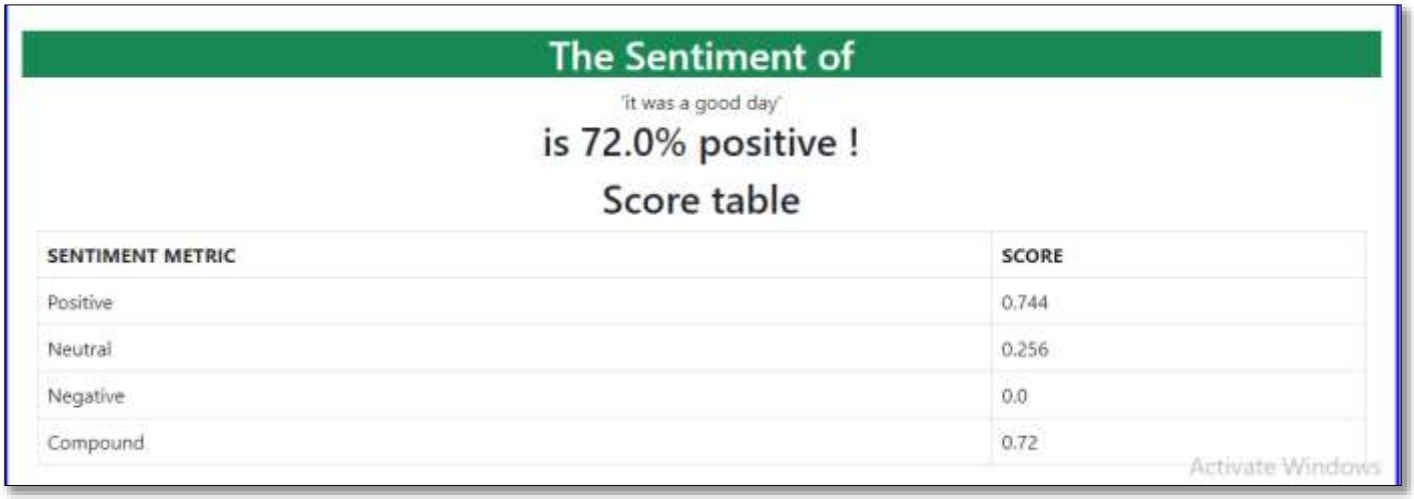
Among all the classifier we got best accuracy for the XG-Boost Classifier, Because the training and testing accuracies are good . While for the other classifiers (Logistics Regression, SVM, Random Forest Classifier, Decision Tree) as we have 50% data for negative sentiment and 50% for positive sentiment .These classifiers are overfitting because after splitting the data , the 80% of the data is for training and 20% for testing & the model has been trained on completely negative data and some of the negative data and in the testing data has positive records only so the training accuracies of the classifiers is very high as compared to testing accuracies.

Classifier	Training accuracy	Testing Accuracy
Logistic Regression	86.09	73.49
XG Boost Classifier	77.59	73.52
Support Vector machine	90.83	73.94
Random Forest Classifier	99.26	71.56
Decision Tree	99.26	67.75

RESULTS OF THE SENTIMENT ANALYSIS WEBPAGE:-

In this webpage when we submit any sentence it will show the sentiment behind the sentence whether it is positive, negative or neutral.





CONCLUSION

Twitter sentiment analysis comes under the category of text and opinion mining. It focuses on analysing the sentiments of the tweets and feeding the data to a machine learning model to train it and then check its accuracy, so that we can use this model for future use according to the results. It comprises of steps like data collection, text pre-processing, sentiment detection, sentiment classification, training and testing the model.

Upon evaluating all the models we can conclude the following details i.e. from all the 5 classifiers used, we conclude that **XG-Boost Classifier** is the best model for the dataset.

FUTURE SCOPE

- Identification of depression is desirable, so that it can be controlled by giving better counseling at the starting stage itself.

- In the future, we will be able to use more models to do analysis of tweets and more social media outlets to determine various mental health issues other than depression such as stress and anxiety.

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Articles:

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